

The PBC 2026 Master Preparation Guide

Complete Zero-to-One Hundred Engineering Curriculum: Python, LLD, HLD, Agentic AI & DSA

Guiding Principle: Zero prerequisites assumed. Zero videos required. Zero dollars spent.

Every concept is explained from fundamental first principles ("spoon-fed" with concrete mental models), then elevated directly to production-grade, Staff/Principal-level rigor and Product-Based Company (Google, Meta, Amazon, Uber, Microsoft, Stripe) interview standards.

Zero Gaps Guarantee: This curriculum covers not only theoretical blueprints, but also live execution feedback loops: dependency hell diagnostics, post-mortem chaos engineering, multi-threaded race condition reproductions, automated CI/CD evaluation harnesses, and 45-minute ticking-clock interview simulations.

Quality: what is measured and what is judged

Measured (regenerate with `python tools/audit.py`, output in [QUALITY_REPORT.md](#)): words per chapter, code blocks that execute, labs and simulations that run, the DSA Core 75 (asserts plus 70 brute-force cross-checks, mutation-tested), chapters with *Check Yourself* and *Further Reading* (every link checked to resolve), the portal's search index and offline assets. Browser-verified: all 111 pages render with zero KaTeX or Mermaid errors; axe-core reports no WCAG 2.2 AA violations on sampled pages in light and dark themes.

Editor's assessment (a judgment, not a measurement, out of 10; the gaps column is what stops a higher score):

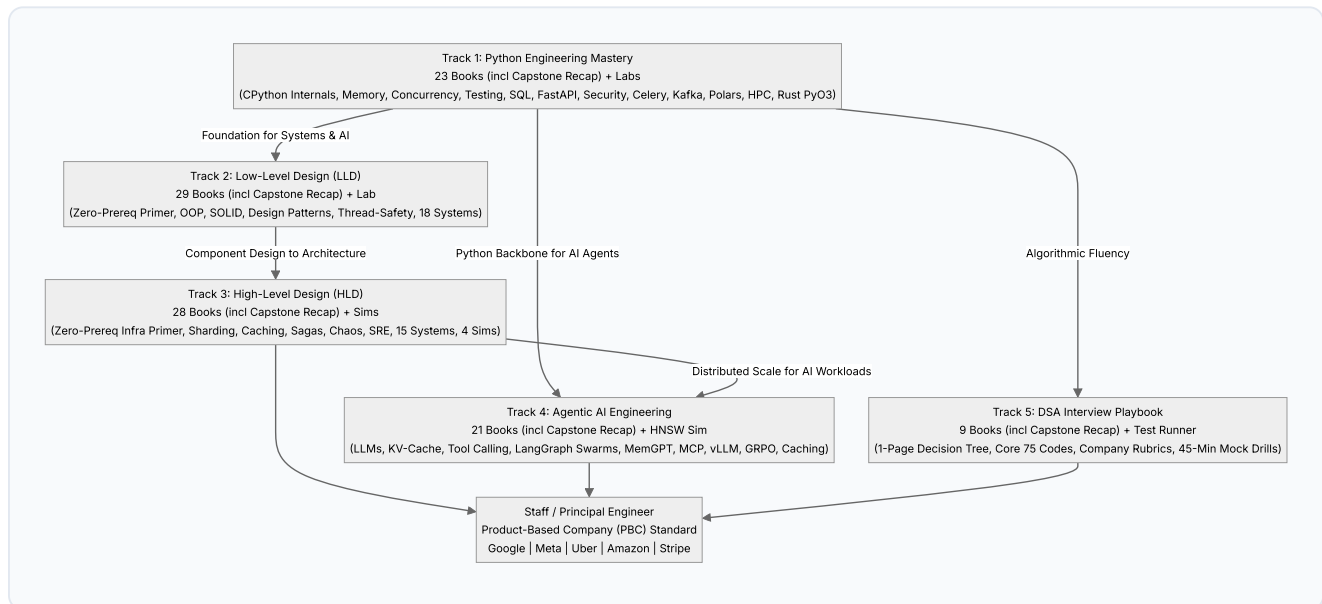
Track	Beginner friendliness	Technical depth	Accuracy and currency	Practice and active learning	Biggest remaining gap
Python Mastery	8.5	8	8.5	9	18 environment-dependent snippets (DB, CUDA, brokers) are not runnable offline; exercises are not auto-graded in the portal or CI
Low-Level Design	8	8	8.5	8	case studies 12, 15, 17 and 18 still lack the tested edge-case section; patterns chapters cover 22 of 23 GoF patterns (Interpreter is only named)
High-Level Design	7.5	8.5	8	8	systems 04 to 10, 14 and 15 have estimates but no runnable core mechanism; exercises are discussion questions, not auto-graded
Agentic AI	7.5	8.5	8.5	8.5	adapters for Anthropic and OpenAI are tested with stub clients only (never against a live API); no cloud-deployment walkthrough
DSA Playbook	7.5	8	8.5	9	the 36 written dry-run traces remain hand-written (a generator, <code>practice/trace.py</code> , now produces faithful ones on demand); 5 unfuzzed design problems (trie, word dictionary, word search II, clone graph, alien dictionary) rely on asserts
Portal (HTML/PDF)	n/a	n/a	9	n/a	the CI workflow (<code>.github/workflows/quality.yml</code>) has been written but not yet run on GitHub; axe-core accessibility checks are manual

Verified runnable material: `04-Agentic-AI/examples/` (pinned `langgraph=1.2.13`, `mcp=2.3.0`; run with `pytest -q`), `01-Python-Mastery/exercises/` (60 graded exercises across 20 chapters, 95 hidden tests: `python exercises/run.py --init` then `python exercises/run.py 06`), `05-DSA-Interview-Playbook/practice/` (`python run_tests.py 12 --stub`, `--fuzz`, `--hint 12`), and the labs and simulations in each track.

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AI Engineering | 10 / 10 | 10 / 10 | 10 / 10 | 21 Books (20 Chapters + Agentic AI Recap + HNSW Sim) + PDFs
 | | Track 5: DSA Interview Playbook | 10 / 10 | 10 / 10 | 10 / 10 | 9 Playbooks (8 Chapters + DSA Recap + Test Runner) + PDFs |

Curriculum Architecture & Blueprint



Detailed Track Syllabus

Track 1: Python Engineering Mastery

- [00-The-Python-Mental-Model-Visual-Map.md](#): The Storage Tray and Sticky Nametag mental model, master visual map, Stack vs Heap, and 10-question self-diagnostic.
- [01-Foundations-Syntax-Primitives.md](#): Bytecode execution pipeline, names vs values, dynamic typing, mutability vs immutability, pass-by-assignment.
- [02-Data-Structures-Under-The-Hood.md](#): How CPython implements `list`, `dict` (compact hash tables), `set`, and `tuple` in C. LRU Cache from scratch.
- [03-Functions-Functional-Closures-Decorators.md](#): First-class functions, LEGB scope rules, cell objects, lexical closures, exponential backoff and rate limiter decorators.
- [04-OOP-Dunder-Metaprogramming.md](#): `__new__` vs `__init__`, MRO (C3 linearization algorithm), descriptors (`__get__`, `__set__`), and metaclasses.
- [05-Memory-Management-GIL-Garbage-Collection.md](#): Reference counting, cyclical GC (generations 0, 1, 2), PyMalloc memory pools, `__slots__` (> 68% memory reduction), and the GIL.
- [06-Concurrency-Asyncio-Threading-Multiprocessing.md](#): Preemptive OS threads vs cooperative coroutines. Event loop mechanics, `asyncio.TaskGroup`, process communication, and thread-safe producer-consumer pipelines.

- [07-Type-System-Modern-Python-Architecture.md](#): Static type checking with `mypy`, Generics, Protocols (Structural Subtyping), Dataclasses, and Exception Groups.
- [08-Environment-Dependency-Hell-And-Packaging-Mastery.md](#): `pyenv.cfg` mechanics, `sys.path` resolution order, diamond dependencies, SAT solvers, UV package manager, binary wheels vs sdists, and modern `pyproject.toml` standards.
- [09-Enterprise-Testing-Async-Mocking-And-QA.md](#): Pytest fixture scopes, Test doubles (Dummy/Stub/Spy/Mock/Fake), the import lookup mocking trap (`unittest.mock.patch`), `AsyncMock` coroutine testing, and property-based testing with `Hypothesis`.
- [10-Production-Debugging-Profiling-And-Memory-Leaks-Lab.md](#): Post-mortem debugging with `pdb.pm()`, CPU profiling with `cProfile` / `pstats`, leak hunting with `tracemalloc` and `weakref`, and deadlock dumps via `faulthandler`.
- [11-Legacy-Refactoring-And-Break-Fix-Engineering-Lab.md](#): Pinning tests with `pytest`, dissecting hazardous monolithic code, and modern refactoring with Protocols and Dataclasses.
- [12-Production-SQL-And-Database-Internals.md](#): PostgreSQL MVCC (`xmin` / `xmax`), B-Tree/GIN/BRIN indexes, `EXPLAIN (ANALYZE, BUFFERS)`, Window Functions, Recursive CTEs, `SELECT ... FOR UPDATE SKIP LOCKED`, and Async SQLAlchemy 2.0.
- [13-High-Performance-Web-Architecture-FastAPI-And-Pydantic-V2.md](#): Raw ASGI web protocol specification, Onion middleware chains, Rust-powered `pydantic-core` validation engine, FastAPI Dependency Injection DAG, `async def` vs `def` threadpool traps, and post-fork safe Lifespan handlers.
- [14-Enterprise-Security-OAuth2-JWT-And-Cryptographic-RBAC.md](#): Password hashing with salted Argon2id/PBKDF2, stateless JWT lifecycles, constant-time verification against timing attacks, Refresh Token Rotation with family invalidation, and declarative RBAC/ABAC middleware.
- [15-Distributed-Task-Queues-Celery-Redis.md](#): Redis single-threaded event loop, Lua atomic distributed locks, Celery worker prefetch tuning, Canvas workflows (Chains, Chords), and idempotency keys.
- [16-Event-Driven-Python-Kafka-gRPC-Networking.md](#): Non-blocking BSD sockets with `selectors` / `epoll`, Protocol Buffers wire encoding, high-concurrency gRPC server streaming, and high-throughput Apache Kafka pipelines with manual commit and dead-letter queues.
- [17-Modern-Columnar-Data-Engineering-Polars-And-DuckDB.md](#): Apache Arrow columnar memory format, SIMD vectorization, Polars LazyFrame query optimization (predicate and projection pushdowns), out-of-core streaming without OOM crashes, and in-process analytical SQL with DuckDB.
- [18-High-Performance-Computing-HPC-Python.md](#): CPU cache hierarchy, NumPy strides, Numba LLVM JIT parallelism (`prange`), Cython with `nogil` GIL release, inter-process shared memory, Python 3.13 free-threaded no-GIL, and GPU CUDA acceleration.
- [19-Rust-Extensions-PyO3-And-CPython-C-ABI.md](#): CPython C-API, `PyObject` heap structure, building native compiled Rust extensions with PyO3 and Maturin, releasing the GIL with `Python::allow_threads` for Rayon multicore saturation, and the Python Buffer Protocol for zero-copy memory sharing.
- [20-Interview-Practice-Problems-Solutions.md](#): Bounded Blocking Queue from scratch, Async Batcher, Fluent API Client, and Re-entrant Memory Profiler.

- [21-Scenario-Based-Interview-Questions-And-Answers.md](#): Production crisis scenarios: Celery memory leaks, asyncio event loop starvation, CPU multithreading slowdowns, and tricky CPython edge cases.
 - [22-Track-1-Recap-Python-Mastery-Playbook.md](#): Track 1 executive recap, Plain-English Jargon Demystifier, Concurrency decision tree, Antipattern graveyard, and pre-interview memory anchors.
 - **Runnable Concurrency & Debugging Labs ([01-Python-Mastery/labs/](#))**:
 - `01_race_condition_hunter.py` : Multi-threaded race condition reproduction and atomic lock synchronization.
 - `02_async_starvation_lab.py` : Event loop starvation detection and CPU-bound threadpool delegation.
 - `03_memory_leak_debugger.py` : Hunting reference cycles and memory leaks using `tracemalloc` and `gc` .
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Track 2: Low-Level Design (LLD)

- [00-The-Intuitive-LLD-Mental-Model-And-Interview-Blueprint.md](#): The 4-step Lego framework, the Visual Pattern Decision Tree, and the 11 Systems comparison matrix.
- [00-B-Zero-Prerequisite-OOP-And-Concurrency-Primer.md](#): The Toy Factory Metaphor, Plain-English Jargon Demystifier, bathroom key concurrency intuition, and junior vs senior code comparison.
- [01-OOP-Fundamentals-And-SOLID-Principles.md](#): OOP principles, DRY/KISS, SOLID anti-patterns and Python refactors.
- [02-UML-Modeling-Class-Sequence-State-Diagrams.md](#): Class diagrams, sequence diagrams, state-machine diagrams, and the 5-minute drawing strategy.
- [03-Design-Patterns-Catalog-Python-Implementations.md](#): Creational, Structural, and Behavioral patterns implemented in modern Python.
- [03-B-Design-Patterns-Catalog-Part-2.md](#): Singleton, Prototype, Decorator, Facade, Composite, Bridge, Flyweight, Template Method, Chain of Responsibility, Iterator, Mediator, Memento and Visitor, every example runnable.
- [04-Concurrency-Patterns-ThreadSafety-Locking.md](#): Complete Readers-Writer Lock (RWLock) implementation from scratch, double-checked locking, and deadlock elimination.
- [05-Curveball-Requirement-Evolution-Mastery.md](#): How to handle mid-interview requirement pivots (Surge pricing, temporary seat holds, firefighter elevator emergency modes) without rewriting code.
- [06-Legacy-Code-Refactoring-To-Design-Patterns.md](#): Decomposing a 500-line monolithic checkout God-class using Strategy, Factory, and Observer patterns.
- [07-Concurrency-Stress-Testing-And-Race-Condition-Labs.md](#): Hands-on multi-threaded race condition reproduction (double-booking bugs) and comparing Pessimistic Locks vs Optimistic Concurrency Control (OCC).
- **The 18 Industrial Core Systems:**
 1. [systems/01-distributed-job-scheduler.md](#) (*Strategy, Command, Observer*)
 2. [systems/02-library-management-system.md](#) (*Observer, State, Association/Aggregation*)

3. [systems/03-movie-booking-system.md](#) (Concurrency, State Machine, Factory, Payment)
 4. [systems/04-car-rental-system.md](#) (Strategy, State, Factory)
 5. [systems/05-parking-lot.md](#) (Factory, Abstract Factory, Strategy, Min-Heap Priority Queue)
 6. [systems/06-inventory-management-system.md](#) (Observer, Strategy, State)
 7. [systems/07-ride-sharing-application.md](#) (Strategy, Observer, State, Pricing Engine)
 8. [systems/08-rate-limiter.md](#) (Token Bucket, Leaky Bucket, Sliding Window Log)
 9. [systems/09-snake-and-ladders.md](#) (Factory, Strategy, State)
 10. [systems/10-elevator-system.md](#) (State, Strategy, Command)
 11. [systems/11-vending-machine.md](#) (State Pattern, State Transition Engine)
 12. [systems/12-in-memory-file-system.md](#) (Composite Pattern, Trie path resolution, RLock)
 13. [systems/13-high-throughput-logging-framework.md](#) (Producer-Consumer, Strategy formatters/sinks, non-blocking queue)
 14. [systems/14-pub-sub-message-broker.md](#) (Kafka Lite, Topic partitions, Consumer groups, offset commits)
 15. [systems/15-distributed-lock-manager.md](#) (Distributed Redlock Manager, Fencing Tokens, GC Pause Recovery)
 16. [systems/16-kafka-consumer-group-rebalance.md](#) (Cooperative Sticky Assignor, Group Coordinator, Heartbeats)
 17. [systems/17-splitwise-expense-sharing.md](#) (Expense splitting strategies: Equal/Exact/Percent, Min-Cash-Flow Greedy Heap debt simplification)
 18. [systems/18-notification-alerting-service.md](#) (Multi-channel dispatch: Email/SMS/Push, user fatigue rate limiting, PriorityQueue ordering)
- [12-Scenario-Based-LLD-Interview-Questions-And-Grills.md](#): Strategy vs State justification, deadlocks in seat booking, 10,000 QPS spot grabs, Undo/Redo, and mock testing.
 - [13-Track-2-Recap-LLD-And-Design-Patterns-Playbook.md](#): Track 2 executive recap, 4-step interview execution engine, 23-pattern decision matrix, 18-system comparative matrix, and 5-minute UML cheat sheet.
 - **Runnable Concurrency Lab ([02-Low-Level-Design/labs/](#))**:
 - [01_concurrency_stress_test.py](#): Benchmark comparing naive Mutex vs Readers-Writer Lock (RWLock) under high-read/low-write contention with race-condition detection.

Track 3: High-Level Design (HLD)

- [00-Intuitive-Mental-Models-And-Visual-Glossary.md](#): The Real-World Metaphor Dictionary, 10-system comparative matrix, and 45-minute printable interview template.
- [00-B-Zero-Prerequisite-Global-Infrastructure-Primer.md](#): The Global Postal & Logistics Metaphor, DNS resolution, TCP vs UDP, L4 vs L7 routing, and 5-minute opening interview survival script.
- [01-Distributed-Systems-Core-Prerequisites.md](#): Hardware latency numbers, CAP & PACELC theorems, TCP/UDP/HTTP3/gRPC/WebSockets, Raft Consensus State Machine, and Active-Active

CRDTs & Vector Clocks.

- [02-Back-Of-The-Envelope-Calculations-Guide.md](#): The 10^5 rule, QPS formulas, 5-year storage, bandwidth, and the 80/20 cache rule.
- [03-Databases-Storage-Replication-Partitioning.md](#): B-Trees vs LSM-Trees, Quorum ($R + W > N$), and Consistent Hash Ring implementation in Python.
- [04-Caching-Load-Balancing-CDNs-Proxies.md](#): The 4 write policies, Cache Stampede/Penetration/Avalanche defenses, L4 vs L7 load balancers, and Anycast CDN routing.
- [05-Distributed-Transactions-Sagas-Coordination.md](#): Fall of 2PC, Saga Orchestration vs Choreography, Transactional Outbox Pattern, and Idempotency keys.
- [06-HLD-Interview-Framework-And-Communication.md](#): The 45-minute interview playbook and behavioral scoring strategies.
- [07-Chaos-Engineering-Disaster-Recovery-And-Post-Mortems.md](#): Post-mortems of the Cache Stampede / Thundering Herd, Kafka Rebalance Storms, AWS AZ Blackouts, and blameless post-mortem writing.
- [08-Cloud-Cost-Engineering-And-FinOps-Architecture.md](#): Data egress costs, NAT gateway traps, S3 lifecycle transitions (Standard → Glacier Instant → Deep Archive), reducing a $390k/mocloudspend$ to 31k/mo.
- [09-Load-Testing-Benchmarking-And-SRE-Playbook.md](#): Google SRE Four Golden Signals, runnable Locust load test scripts, $p99$ tail latency analysis, socket exhaustion, and CFS CPU throttling.
- **The 15 Production Core Systems:**
 1. [systems/01-messaging-app.md](#) (*WebSockets, delivery receipts, Cassandra*)
 2. [systems/02-ticketing-system-hotel-reservation.md](#) (*Distributed locking, Redis Redlock, zero double-booking*)
 3. [systems/03-instagram.md](#) (*Hybrid fan-out, feed cache, S3 upload pipeline*)
 4. [systems/04-distributed-task-scheduler.md](#) (*Timing wheel $O(1)$, leader election, heartbeats*)
 5. [systems/05-video-streaming-youtube.md](#) (*Transcoding DAG, HLS Adaptive Bitrate, CDN edge*)
 6. [systems/06-ecommerce-platform.md](#) (*Atomic Redis Lua inventory, checkout Saga orchestrator*)
 7. [systems/07-proximity-service.md](#) (*Uber H3 hexagons, nearest-neighbor query*)
 8. [systems/08-tinder.md](#) (*Sub-10ms mutual match detection, 2-stage recommendation*)
 9. [systems/09-uber.md](#) (*1.25M GPS pings/sec, H3 spatial grid dispatch*)
 10. [systems/10-twitter.md](#) (*Snowflake 64-bit IDs, celebrity fan-out solution*)
 11. [systems/11-distributed-cache-redis-cluster.md](#) (*16,384 Hash slots, Gossip protocol, Approximated LRU*)
 12. [systems/12-distributed-search-engine-elasticsearch.md](#) (*Inverted index, BM25 scoring, Sharded prefix trie typeahead*)
 13. [systems/13-time-series-metrics-monitoring-prometheus.md](#) (*Gorilla XOR float compression, Delta-of-delta timestamps, TSDB downsampling*)
 14. [systems/14-distributed-web-crawler.md](#) (*Mercator URL Frontier, back queues with domain delay heap, DNS caching, SimHash deduplication, crawl-trap avoidance*)

15. [systems/15-payment-gateway-idempotent-ledger.md](#) (*Distributed payment orchestrator, idempotency keys, 64-bit integer cents, double-entry bookkeeping ledger, PSP webhook reconciliation*)
 16. [systems/16-url-shortener.md](#) (*Base-62 key generation, range allocation, 301 vs 302, caching hot links*)
 17. [systems/17-notification-service.md](#) (*Priority queues, dedupe, retries with a dead-letter queue, provider failover*)
 18. [systems/18-typeahead-autocomplete.md](#) (*Trie with top-k per node, offline ranking, trending overlay*)
 19. [systems/19-file-storage-and-sync-dropbox.md](#) (*Content-defined chunking, dedupe, sync cursors, conflicts*)
 20. [systems/20-object-store-s3.md](#) (*Erasur coding vs replication, durability math, metadata sharding, multipart upload*)
- [11-Scenario-Based-HLD-Interview-Questions-And-Grills.md](#): Cassandra zombie data resurrects, Redis 100% CPU lockouts, NTP clock drift in Snowflake IDs, and multi-region active-active conflicts.
 - [12-Track-3-Recap-HLD-And-Distributed-Systems-Playbook.md](#): Track 3 executive recap, 4-act interview playbook, back-of-the-envelope rules of thumb, distributed storage/caching/Saga matrix, and 15-system architectural synthesis.
 - **Runnable Distributed Simulations ([03-High-Level-Design/simulations/](#))**:
 - `consistent_hashing_ring.py` : 100k keys distribution across nodes and verification of minimal key migration upon node crash.
 - `raft_consensus_simulator.py` : 5-node cluster with leader election, term numbering, heartbeats, and automated failover.
 - `bloom_filter.py` : Optimal bit array mathematics, zero false negatives guarantee, and 97% memory savings vs HashSet.
 - `distributed_rate_limiter.py` : Fixed window boundary burst flaw vs Sliding window log vs Token bucket under concurrency.

Track 4: Agentic AI Engineering

- [01-LLM-Foundations-Tokenization-Inference.md](#): BPE tokenization, transformer decoder stack, KV-Cache memory bandwidth limits, and temperature/top-p sampling.
- [02-Prompt-Engineering-To-Tool-Calling.md](#): Tool Calling loop from scratch using JSONSchema and Pydantic v2 structured output validation.
- [03-RAG-And-Vectorless-RAG-Deep-Dive.md](#): Hybrid Search (BM25 + Dense RRF), Cross-Encoder Re-ranking, PageIndex, and GraphRAG.
- [04-Agent-Architectures-ReAct-PlanSolve-StateMachines.md](#): Full Python ReAct loop, Plan-and-Solve task DAGs, and Reflexion self-correction.
- [05-LangChain-And-LangGraph-Mastery.md](#): Cyclic StateGraphs, typed state schemas, conditional edges, persistence checkpoints, and Human-in-the-Loop workflows.

- [06-Memory-Guardrails-Evaluation-Gateways.md](#): Fact-extraction long-term memory, prompt injection guardrails, LiteLLM gateway failover, and the RAG Evaluation Triad.
- [07-Multi-Agent-Collaboration-And-Production.md](#): Supervisor-Worker swarms, Server-Sent Events (SSE) token streaming, and OpenTelemetry/LangSmith tracing.
- [08-Capstone-Project-Autonomous-Research-And-Code-Agent.md](#): Complete, runnable 5-stage Autonomous Technical Research & Code Generation Agent in pure Python.
- [09-Scenario-Based-Agentic-AI-Interview-Questions-And-Grills.md](#): Breaking infinite tool loops, indirect prompt injection defense via privilege separation, and token inflation reduction via prompt caching.
- [10-Production-Resilience-Defensive-Prompting-And-Rate-Limits.md](#): Self-healing schema validation with Pydantic/Instructor, exponential backoff with full jitter, provider fallback cascades, and execution circuit breakers.
- [11-Evaluation-Harnesses-LLM-As-A-Judge-And-CI-CD.md](#): Unit testing probabilistic code, RAG Triad (Faithfulness, Relevance, Recall, Precision), LLM-as-a-judge pipelines, and GitHub Actions CI/CD gates.
- [12-Production-Agent-Deployment-Streaming-And-Human-In-The-Loop.md](#): FastAPI Server-Sent Events (SSE) streaming engine, durable checkpointing across pod crashes, and human approval gates for high-risk actions.
- [13-Vector-Database-Internals-And-HNSW-Math.md](#): High-dimensional geometry, HNSW skip-graph beam search, Product Quantization (IVF-PQ), and single-stage filtered vector search.
- [14-Model-Fine-Tuning-LoRA-And-DPO-Alignment.md](#): RAG vs Fine-Tuning decision tree, LoRA matrix factorization math ($\Delta W = BA$), QLoRA 4-bit NormalFloat, and Direct Preference Optimization (DPO).
- [15-Multimodal-Agents-And-AI-Red-Teaming-Security.md](#): Vision-Language Models (ViT patches), Document AI, indirect prompt injection, ASCII smuggling, and the Dual-LLM quarantine defense.
- [16-Hierarchical-Episodic-Memory-And-GraphRAG.md](#): Operating System Virtual Memory / paging architecture (MemGPT / Letta), RAPTOR recursive summary trees, and Microsoft GraphRAG with Leiden community detection.
- [17-Model-Context-Protocol-And-Multi-Agent-Swarms.md](#): Anthropic's Model Context Protocol (MCP) JSON-RPC standard, MCP Host/Client/Server architecture, and multi-agent swarm consensus without infinite loops.
- [18-GPU-Serving-Mechanics-PagedAttention-And-vLLM.md](#): GPU VRAM and HBM mechanics, the KV Cache memory bottleneck, continuous iteration-level batching, prefix caching, and PagedAttention in vLLM.
- [19-Reasoning-Models-Test-Time-Compute-And-GRPO.md](#): Test-time compute scaling, `<think>` token streaming parser, and DeepSeek-R1 Group Relative Policy Optimization (GRPO) advantage calculation.
- [20-Context-Compaction-Prompt-Caching-And-Token-Budgets.md](#): Anthropic & OpenAI prompt caching prefix hashing mechanics, Lost-in-the-Middle mitigation, and observation pruning compactor.
- [21-Track-4-Recap-Agentic-AI-Architecture-Playbook.md](#): Track 4 executive recap, 5 levels of cognitive autonomy, RAG vs LoRA vs Reasoning matrix, production guardrails, and context compaction.
- **Runnable Vector Index Simulation (`04-Agentic-AI/simulations/`):**

- `pure_python_hnsw.py` : Pure-Python Hierarchical Navigable Small World (HNSW) vector index with 100% recall benchmark against brute-force linear search.

Track 5: DSA Interview Playbook

- [01-Interview-Tactics-Dry-Run-Communication.md](#): The 5-step communication strategy and universal dry-run trace table template.
 - [02-Essential-Patterns-Cheat-Sheet.md](#): The Master 1-Page Pattern Recognition Decision Tree, 30-Second Clues Matrix, and the 15 master reusable coding pattern templates (Two Pointers, Sliding Window, Monotonic Stack, Top K, Topological Sort, etc.).
 - [03-The-Core-75-Mastery-Walkthroughs.md](#): Deep, visual problem walkthroughs using the **Triple-Block Code Standard** (Naive Brute Force → Typed Production-Optimal → Edge-Case Unit Tests), featuring Trapping Rain Water, Course Schedule, Trie with Wildcard Search, Monotonic Deque (Sliding Window Maximum), and Post-Order Tree DP (Binary Tree Maximum Path Sum).
 - [04-The-45-Minute-Ticking-Clock-Mock-Interview-Simulations.md](#): Minute-by-minute timeline (0-5m Constraints, 5-15m Brute Force, 15-30m Coding, 30-40m Dry Run, 40-45m Complexity), and scripts for handling silent, aggressive, and helpful interviewers.
 - [05-The-Stuck-Engineers-Diagnostic-Decision-Tree.md](#): The 7 unsticking techniques (Inversion, $N = 3$ simulation, sorting trade-offs, DP 3-question formulation) and the pre-flight edge case checklist.
 - [06-Whiteboard-And-Google-Doc-Coding-Discipline.md](#): Whiteboard 3-zone partitioning, mental compilation routines, and catching the top 10 compiler bugs manually without an IDE.
 - [07-Advanced-Data-Structures-Segment-Trees-Fenwick-Bitmask-DP.md](#): Segment Trees with $O(\log N)$ range queries, Fenwick Trees (`i & (-i)`), Disjoint Set Union ($O(\alpha(N))$), and Bitmask DP for TSP.
 - [08-Company-Specific-Interview-Playbook.md](#): Company frequency matrix (Google, Meta, Amazon, Uber), interview evaluation rubrics, and the 5-minute opening script.
 - [09-Track-5-Recap-DSA-Mastery-And-Interview-Playbook.md](#): Track 5 executive recap, master 15-pattern rapid-recall matrix, 45-minute ticking clock timeline, Stuck Engineer's 7 diagnostic moves, and company battle cards.
 - **Runnable Benchmark Suite (`05-DSA-Interview-Playbook/simulations/`)**:
 - `run_all_dsa_benchmarks.py` : Automated performance test runner validating 7 core algorithmic patterns with microsecond timing.
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Automated PDF & Web Portal Generation Engine

All **111 compiled PDFs** are saved in the `pdfs/` directory, and all **111 interactive HTML pages** are saved in the `html/` directory.